

SOFTWARE REQUIREMENTS SPECIFICATION

For 3D Interactive MAP



KOLAHPUR INSTITUTE OF TECHNOLOGY, KOLHAPUR

- Class :- SY
- Div :- B
- Course :- Mini Project -I
- Team Members :-
  - Adarsh Gurav
  - Sanket Magdum
  - Pratik Patil
  - Anusha Patil

Table of Contents

1. Introduction 1,2

1.1 Purpose 1

1.2 Document Conventions 1

1.3 Intended Audience and Reading Suggestions 1

1.4 Product Scope 2

1.5 References 2

1.6 Benefits 2

2. Overall Description 3,4,5

2.1 Product Perspective 3

2.2 Product Functions 3

2.3 User Classes and Characteristics 3

2.4 Operating Environment 3

2.5 Design and Implementation Constraints 4

2.6 Assumptions and Dependencies 4

2.7 User Interaction Flow 5

3. External Interface Requirements 6,7,8

3.1 User Interfaces 6

3.2 Hardware Interfaces 6

3.3 Software Interfaces 7

3.4 Communications Interfaces 7

3.5 Search Function Flow 8

4. System Features 9,10,11

3D Map Display 9

4.1 Map Navigation 9

4.2 Multilanguage content Support 9

4.3 Markers and Info Popups.....9
4.4 Search Function 10
4.5 UI Control.....10
4.6 Responsive Design.....10
4.7 Performance.....11

5. Other Nonfunctional Requirements……12,13

5.1 Performance Requirements.....12
5.2 Usability.....12
5.3 Reliability 12
5.4 Portability 12
5.5 Maintainability.....13
5.6 Scalability 13

1. Introduction :-

1.1 Purpose

The purpose of this project is to design and develop a 3D Interactive Map using web
Technologies (HTML, CSS, JavaScript) and 3D rendering libraries like Three.js. The
system will allow users to explore a geographical area with features such as zoom, pan,
hover effects, and clickable regions with information pop-ups.

1.2 Document Conventions

- **Bold**: Used for section titles, feature names, and emphasis on key terms.
- *Italics*: Used for UI labels, buttons, panels, and options visible in the interface.
- FR – Functional Requirement (e.g., FR1: Display the 3D map on page load)
- NFR – Non-Functional Requirement (e.g., NFR1: The map should load within 5 seconds)

1.3 Intended Audience and Reading Suggestions

- Frontend Developers
- Tourists / Students / General users
- Project evaluators or faculty members

1.4 Product Scope

This project aims to provide users with an engaging visual experience for understanding.
locations, routes, or landmarks. It can be used for:

- Educational purposes (e.g., learning geography)
- Tourism websites
- City planning visualizations

1.5 References

- HTML5, CSS3
- JavaScript (ES6+)
- Three.js
- WebGL
- JSON for location data (if used)

1.5 Benefits

- Enhanced User Experience
- Multilingual Accessibility
- Audio Narration for Inclusivity
- Educational Value
- Cultural Preservation
- Smart City Integration

2. Overall Description :-

2.1 Product Perspective

This is a **Standalone Web Application**. It does not depend on backend services or databases for basic functioning but can later be extended.

2.2 Product Functions

- Display a 3D map of a specified area.
- Allow users to zoom, rotate, and pan the map.
- Interactive hotspots or markers to display information.
- bar to find a location or landmark.
- Tooltip or popup on hover/click.

2.3 User Classes and Characteristics

- Visitor: Can navigate and explore the map.
- Admin: Can update marker data.

2.4 Operating Environment

The **3D Interactive Map** is a frontend web application and will run within a browser environment. It does not require any server-side or backend components for its basic operations.

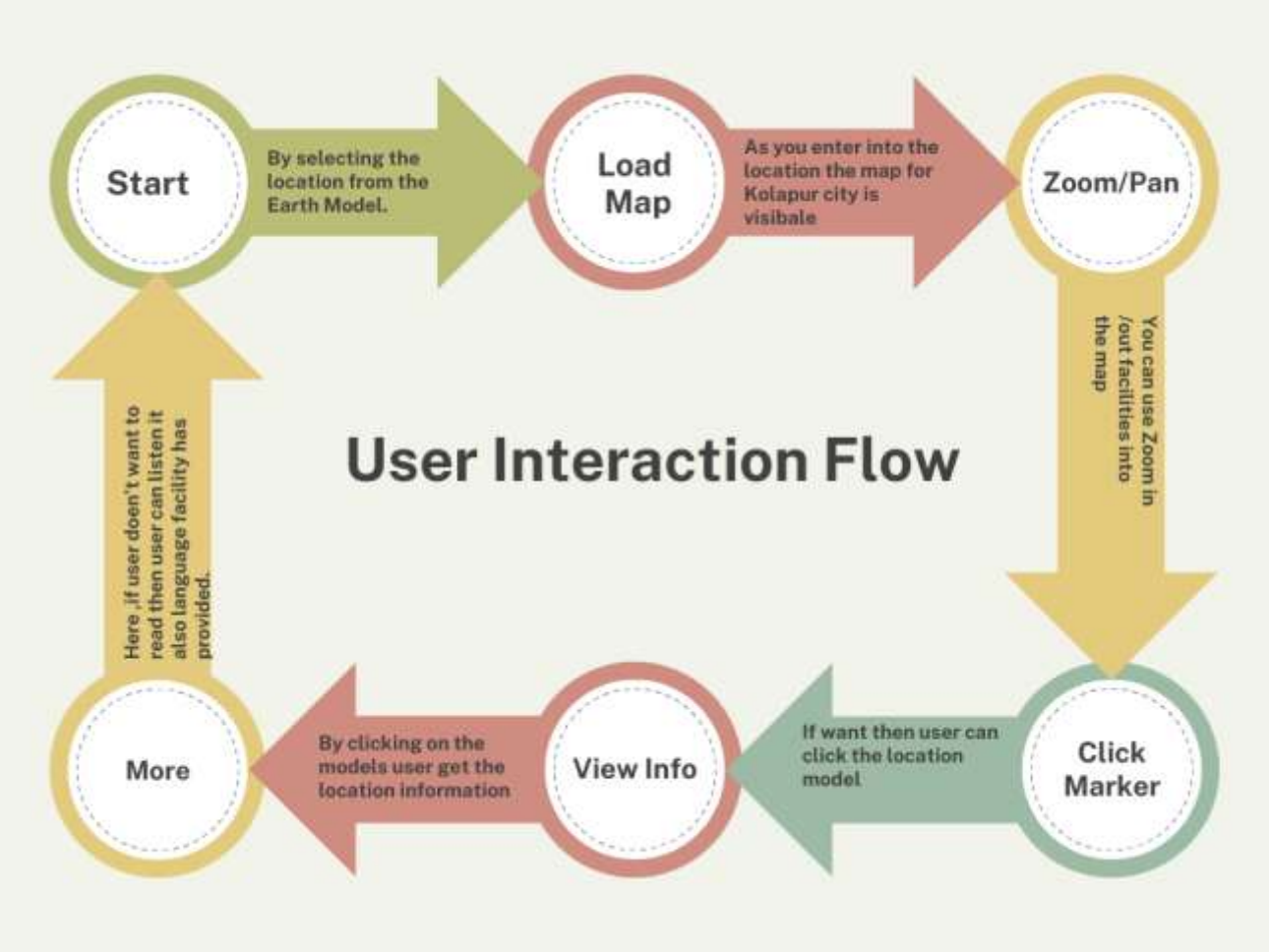
2.5 Design and Implementation Constraints

- Runs only on modern browsers supporting WebGL.
- Needs a stable internet connection to load external libraries or map textures.
- Designed only for desktop view (mobile responsiveness is optional).

2.6 Assumptions and Dependencies

- Internet connection is available.
- Three.js and other scripts are correctly linked.
- All resources (models, textures) are locally stored or accessible via CDN.

2.7 User Interaction Flow



3. External Interface Requirements :-

3.1 User Interfaces

The system shall provide a browser-based interface with:

- A 3D map viewer area.
- Zoom, rotate, and pan controls (mouse or on-screen buttons).
- Clickable markers that show location-specific information in popups.
- A search bar to find and navigate to specific locations.

The UI will be built using:

- HTML5 for structure
- CSS3 for styling
- JavaScript (ES6) for logic and interactivity

3.2 Hardware Interfaces

- No special hardware is required.

Users need a device with:

- Mouse or touch input
- Graphics card supporting WebGL
- Standard screen resolution (minimum 1366×768)

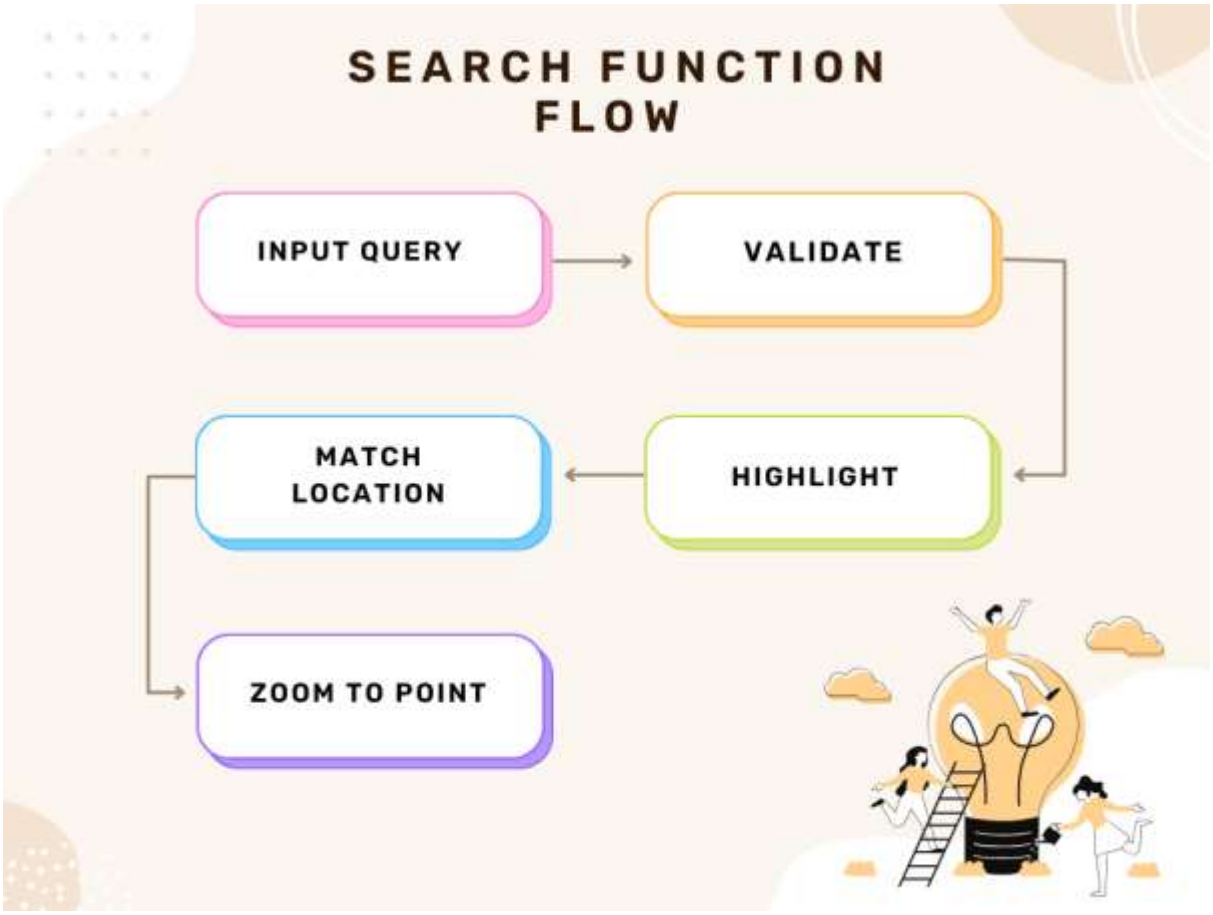
3.3 Software Interfaces

- Runs on modern web browsers (latest versions of Chrome, Firefox, Edge, Safari).
- Uses the following libraries:
    - Three.js for 3D rendering
    - OrbitControls.js for camera movement
    - Optional: Mapbox or other APIs for enhanced features
- File formats used:.js, .json, .html, .css

3.4 Communications Interfaces

- Local Mode: Works entirely on the client side if assets are bundled.
- Online Mode (Optional): If external assets or APIs are used, an internet connection is required.
- No real-time communication or backend APIs are used in the base version.

3.5 Search Function Flow



4. System Features :-

4.1 3D Map Display

- Renders a 3D map using Three.js and WebGL.
- Users can view terrain or city layouts in 3D.
- User can see detail information about the location.

4.2 Map Navigation

- Mouse controls like:
  i. Zoom In, zoom out
  ii. Rotate
  iii. Pan
- Reset view button included.

4.3 Multilingual Content Support

- **Languages Supported**: Marathi IN, Hindi IN, English 🌍
- **Dynamic Language Switching**: Users can toggle language via a dropdown or settings menu.
- **Localized Descriptions**: Each location has metadata (name, history, significance) in all three languages.
- **Unicode Compliance**: Ensures proper rendering of Devanagari script for Marathi and Hindi.

4.4 Markers and Info Popups

- Clickable markers on each landmark or point of interest.
- Popups display:
    - Name of the place
    - Description in selected language
    - Audio play button
    - Images or videos
    - External links (e.g., tourism site, Wikipedia)
    - Show tooltips on hover and popups on click.

4.5 Search Function

- Users can search locations.
- Highlights and zooms to the searched point.

4.6 UI Controls

- Zoom in/out buttons.
- Optional: Layer toggle or legend panel.
- **Responsive Design**: Works on desktop, tablet, and mobile.
- **Language Toggle**: Prominently placed for easy switching.
- **Accessibility Mode**: High contrast, larger fonts, and keyboard navigation.
- **Map Legend**: Explains icons and categories in selected language.

4.7 Responsive Design (Optional)

- Basic support for tablets and mobile screens.

4.8 Performance

- Fast map loading and smooth interaction.
- Lazy loading of 3D assets for faster performance.
- Caching of audio files for smoother playback.
- CDN support for global access.

5. Other Nonfunctional Requirements :-

5.1 Performance Requirements

- The map should load within 5 seconds on a standard internet connection.
- Frame rate should remain smooth (30+ FPS) during interactions.

5.2 Usability

- The UI must be easy to navigate for first-time users.
- Tooltip/help icons should explain basic controls (zoom, rotate, click).

5.3 Reliability

- The application should function consistently across supported browsers.
- It must handle missing data or broken links gracefully (e.g., show a message).

5.4 Portability

- Should work on all modern browsers (Chrome, Firefox, Edge, Safari).
- No installation required – runs directly in the browser.

5.5 Maintainability

- Code should be modular and well-commented for easy updates.
- External libraries (e.g., Three.js) must be used with clear versioning.

5.6 Scalability (Optional)

- The system should be easy to scale for larger maps or additional features (e.g., layers, routes) in future updates.